Machine Learning Analysis of HERA Data

Florini Pettiss, Dr. Christina Love

Drexel University, Department of Physics

Background

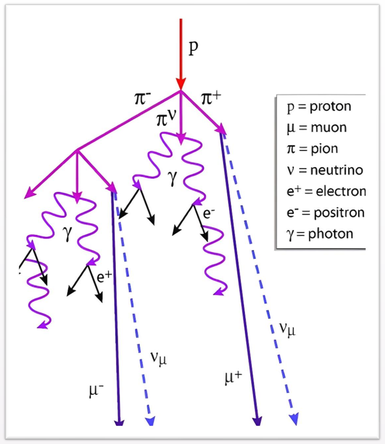


Figure 1: Diagram of a particle shower

Cosmic Rays

- High energy charged particles
- Can create particle showers when they interact with the atmosphere (Gaisser, 2016)
  - Can produce a variety of particles such as muons

HERA

- Launches high-altitude balloons with low-cost, light weight particle detectors (called CosmicWatch detctors)
- Can reach 100,000 ft, or around 30 km
- Data measured by these and other devices can be used/analyzed to learn about comic rays

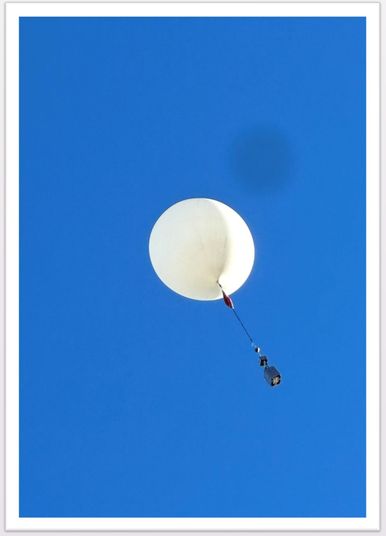


Figure 2: High-altitude balloon (HAB) launched by HERA midair

Devices

CosmicWatch Detector

- Uses a silicon photomultiplier (SiPM) and a plastic scintillator (Axani, 2019)
- Responsible for counts per second data



Figure 3: Preparations to launch a HAB

Adafruit CLUE Datalogger

- Responsible for temperature, pressure, acceleration, magnetic field, latitude, longitude, and altitude data

Methods

This was the first implementation of machine learning with HERA data. Three individual flights were chosen for the completeness of their data. The flights chosen were from March 28, 2026; October 4, 2025; and June 1, 2025.

- Features were standardized using scikitlearn's StandardScaler,
- Acceleration and magnetic field data initially came with x, y, and z components, however, to reduce the amount of features and to reduce the effect of magnetometer calibration issues, the magnitude for both of these measurements was calculated and used as a feature instead.

Regression Models Used

Decision tree based models were chosen because of their easily accessible feature importances attribute. The two models were chosen based on their differences. Random forests do not overfit from adding more trees (Breiman, 2001). Therefore, a random forest model provides a strong contrast to the gradient boosting model, which is more prone to overfitting.
When fitted, for all flights the r^2 score for both models was between 0.98 and 0.99.

Data/Results

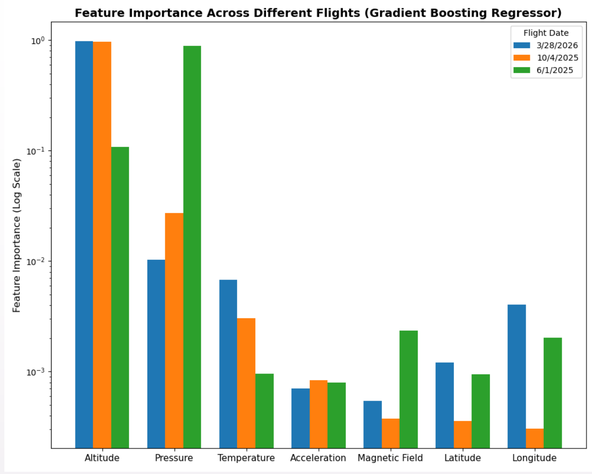


Figure 4: Feature importances by flight using Gradient Boosting

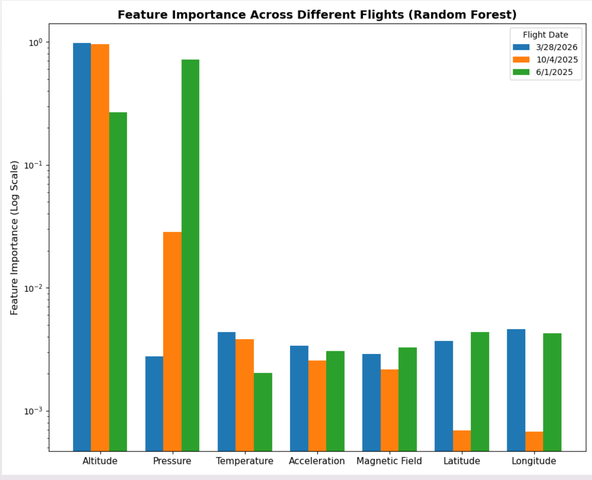


Figure 5: Feature importances by flight using Random Forest

The left-adjacent figures (Figures 4 and 5) show a comparison of feature importance for two different regression models.

- The plots are on a log scale, as the least important had an extremely low importance relative to the most important features
- For two of the three flights, altitude was the most important feature, with a feature importance >.9.
- However, the June 1 flight had drastically different results, with pressure as the most important feature.

Conclusions

- Regression-based machine learning is a possible approach to analyzing HAB data
- Regression models provide an indication of most important features in predicting counts per second, specifically decision tree based models
- More flight data is needed to determine how consistent feature importance is across different flights

Future Work

- Proper calibration of magnetometers to ensure more accurate data

The importance of altitude makes it hard to evaluate the importance of other features such as magnetic field, so possible future work could include

- Using data only gathered from the ground, where altitude is constant
- Using a specific range of high altitude for less of a range in altitude overall
- Fitting a model to the relationship between altitude and counts per second, then analyzing the residuals

References

Axani S. N. G. (2019). The Physics Behind the CosmicWatch Desktop Muon Detectors. Master's thesis, Massachusetts Institute of Technology. arXivpreprint arXiv:1908.00146.

Breiman, L. (2001). Random Forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Gaisser, T. K., Engel, R., & Resconi, E. (2016). Cosmic Rays and Particle Physics. Cambridge University Press.